

HAOYU ZHANG

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Harbin Institute of Technology (Shenzhen)

EDUCATION

- **Harbin Institute of Technology** Sep 2023 - Now
PhD in Electronic Information, School of Computer Science and Technology, GPA: 3.3/4.0
Shenzhen, China
Advisor: Prof. [Liqiang Nie](#), Prof. [Yaowei Wang](#), and Prof. [Meng Liu](#)
- **Shandong University** Sep 2020 - Jun 2023
Master in Computer Technology, School of Computer Science and Technology, GPA: 91.6/100.0
Qingdao, China
Advisor: Prof. [Liqiang Nie](#) and Prof. [Meng Liu](#)
- **Shandong University of Science and Technology** Sep 2016 - Jun 2020
Bachelor in Digital Media Technology, School of Computer Science and Engineering, GPA: 90.7/100.0
Qingdao, China
Recommended for admission to graduate study without entrance examination

RESEARCH INTERESTS

- **Spatial Intelligence:** developing spatial reasoning and embodied perception capabilities for indoor environments and low-altitude UAV scenarios.
- **Egocentric Vision:** understanding first-person perspectives by modeling temporal dynamics, localizing events, and anticipating future actions.
- **Multimodal Retrieval:** learning unified multimodal representations to support dialogue-based retrieval, personalized recommendation, and person re-identification.

FUNDINGS

- **NSFC Young Student Basic Research Program (Doctoral Students), Project Lead** 2026
- **HIT Idea Fund Program (Young Students), Project Lead** 2025

PUBLICATIONS

C=CONFERENCE, J=JOURNAL, P=PREPRINT

- [P.4] From Where to How: Continuous 4D Interaction Forecasting from Egocentric Video 
*Chu, Q., **Zhang, H.**, Liu, M., Shi, H., Jiang, D., Nie, L.*
arXiv Preprint, 2026
- [P.3] SpatialUAV: Benchmarking Spatial Intelligence for Low-Altitude UAV Perception, Collaboration, and Motion 
***Zhang, H.**, Liu, M., Xiang, Q., Wang, K., Wang, Y., Nie, L.*
arXiv Preprint, 2026
- [P.2] UniCVR: From Alignment to Reranking for Unified Zero-Shot Composed Visual Retrieval 
*Wen, H., Song, X., **Zhang, H.**, Guan, W., Zhao, X., Nie, L.*
arXiv Preprint, 2026
- [P.1] TINA+: Probing Residual Visual Knowledge in Unlearned Diffusion Models via Diffusion-Consistent Text-Free Inversion 
*Xiang, Q., Zhang, M., Wang, K., **Zhang, H.**, Hou, J., Nie, L.*
arXiv Preprint, 2026
- [C.9] Beyond Waypoints: A Trajectory-Centric Waypointing Paradigm for Vision-Language Navigation 
*Shi, H., Deng, X., **Zhang, H.**, Chu, Q., Wang, Y., Nie, L.*
Regular Paper in **ACM MM** 2026
- [J.3] SpaceEra++: A Unified Framework Towards 3D Spatial Reasoning in Video 
*Guan, W., **Zhang, H.**, Liu, M., Xiang, Q., Wang, Y., Nie, L.*
Regular Paper in **IEEE TPAMI** 2026 (Co-mentor as first author)
- [J.2] Uncovering Hidden Connections: Iterative Search and Reasoning for Video-grounded Dialog 
***Zhang, H.**, Liu, M., Feng, Y., Wang, Y., Guan, W., Nie, L.*
Regular Paper in **ACM TOIS** 2026

[C.8]	TINA: Text-Free Inversion Attack for Unlearned Text-to-Image Diffusion Models <i>Xiang, Q., Zhang, M., Zhang, H., Wang, K., Hou, J., Nie, L.</i> Regular Paper in CVPR 2026	
[C.7]	Exo2Ego: Exocentric Knowledge Guided MLLM for Egocentric Video Understanding <i>Zhang, H., Chu, Q., Liu, M., Shi, H., Wang, Y., Nie, L.</i> Regular Paper in AAAI 2026	
[C.6]	Intention-Guided Cognitive Reasoning for Egocentric Long-Term Action Anticipation <i>Chu, Q., Zhang, H., Liu, M., Feng, Y., Shi, H., Nie, L.</i> Regular Paper in AAAI 2026	
[C.5]	Spatial Understanding from Videos: Structured Prompts Meet Simulation Data <i>Zhang, H., Liu, M., Li, Z., Wen, H., Guan, W., Wang, Y., Nie, L.</i> Spotlight Paper (3%) in NeurIPS 2025	
[C.4]	Object-Shot Enhanced Grounding Network for Egocentric Video <i>Feng, Y., Zhang, H., Liu, M., Guan, W., Nie, L.</i> Regular Paper in CVPR 2025	
[C.3]	Multi-Factor Adaptive Vision Selection for Egocentric Video Question Answering <i>Zhang, H., Liu, M., Liu, Z., Song, X., Wang, Y., Nie, L.</i> Regular Paper in ICML 2024	
[J.1]	Attribute-guided Collaborative Learning for Partial Person Re-identification <i>Zhang, H., Liu, M., Li, Y., Yan, M., Gao, Z., Chang, X., Nie, L.</i> Regular Paper in IEEE TPAMI 2023	
[C.2]	Bi-directional Heterogeneous Graph Hashing towards Efficient Outfit Recommendation <i>Guan, W., Song, X., Zhang, H., Liu, M., Yeh, C.-H., Chang, X.</i> Oral Paper in ACM MM 2022	
[C.1]	Multimodal Dialog System: Relational Graph-based Context-aware Question Understanding <i>Zhang, H., Liu, M., Gao, Z., Lei, X., Wang, Y., Nie, L.</i> Regular Paper in ACM MM 2021	

INTERN EXPERIENCE

• Academic Intern, Community Science, Kuaishou Technology <i>Multimodal Large Language Model</i> Advised by Fang Yang, Bin Wen	<i>Jul 2024 - Jul 2025</i>
• Academic Intern, Alibaba Cloud Intelligence, Alibaba Group <i>Partial Person Re-identification</i> Advised by Jian Gu, Jinpeng Xia	<i>Wed 2022 - Jul 2022</i>

HONORS AND AWARDS

• Best Oral Presentation Award at PolyU COMP-HIT (Shenzhen) CST Research Student Conference	<i>2026</i>
• Chunhui Innovation Achievement Award (HIT's top science and technology award)	<i>2026</i>
• Heilongjiang Provincial Model Student	<i>2025</i>
• Tat-Seng Chua Scholarship	<i>2025</i>
• NeurIPS Scholar Award	<i>2025</i>
• INSUN Education Scholarship	<i>2024</i>
• ICML Travel Award	<i>2024</i>
• Outstanding Graduates of Shandong University	<i>2023</i>
• National Scholarship	<i>2021</i>
• ACM MM Student Travel Award	<i>2021</i>
• Outstanding Graduates of Shandong Province	<i>2020</i>
• Top 10 Outstanding Students in the School of Computer Science and Engineering	<i>2019</i>

COMPETITIONS

• CVPR EgoVis Workshop, Ego4D Challenge, Champion × 3 <i>(Natural Language Query, Goal Step, and Short-Term Object Interaction Anticipation tracks)</i>	<i>2026</i>
• CVPR EgoVis Workshop, EPIC-Kitchens Challenge, Champion <i>(Action Anticipation track)</i>	<i>2026</i>
• CVPR EgoVis Workshop, Ego4D Challenge, Champion × 4 <i>(Natural Language Query, Goal Step, Moment Queries, and Long-Term Action Anticipation tracks)</i>	<i>2025</i>
• CVPR EgoVis Workshop, Ego4D Challenge, Champion <i>(EgoSchema track)</i>	<i>2024</i>

ACADEMIC SERVICES

- **Conference Reviewer:** CVPR, ICCV, NeurIPS, ICML, ICLR, SIGIR, ACM MM, AAAI, IJCAI, ICMR, PRCV, ISCAS, ECAI
- **Journal Reviewer:** IEEE TPAMI, IEEE TIP, IEEE TKDE, IEEE TCSVT, IEEE TMM, Information Sciences
- **Volunteer:** ICML, China 3DV